

# Mathematics - Pigeon Hole Principle

**Time Allowed : 60 Minutes** \_\_\_\_\_

**Maximum Marks : 40**

Please read the instructions carefully. You will be allotted 5 minutes specifically for this purpose.

## Instructions

### A. General

1. Blank papers, clipboards, log tables, slide rules, calculators, cellular phones, pagers, and electronic gadgets in any form are not allowed.
2. Do not break the seals of the question-paper booklet before instructed to do so by the invigilators.

### B. Question paper format and Marking Scheme :

1. Paper contains 4 questions. Each question in this section contains 10 marks.

- Q1:** Prove that if seven distinct numbers are selected from  $\{1, 2, \dots, 11\}$ , then some two of these numbers sum to 12.
- Q2:** Prove that if five points are selected from the interior of a  $1 \times 1$  square, then there are two points whose distance is less than  $\sqrt{2}/2$ .
- Q3:** Prove that if 11 integers are selected from among  $\{1, 2, \dots, 20\}$ , then the selection includes integer  $a$  and  $b$  such that  $a - b = 2$ .
- Q4:** Prove that if 11 integers are selected from among  $\{1, 2, \dots, 20\}$ , then the selection includes integers  $a$  and  $b$  such that  $b = a + 1$ .